lysa

Microscopy Image Viewer & Analysis Tool

User Manual

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1. Quick Introduction

Lysa is a browser-based viewer for microscopy image data. It runs locally on your machine: you start a small Python server, open `http://localhost:8050`, and your data never leaves your computer.

What makes it different from opening a TIFF in Preview or running ImageJ:

- **It keeps 16-bit data 16-bit.** 16-bit fluorescence images don't get force-converted to 8-bit on load. Window/level adjustments use the native bit depth.
- **It opens Leica LIF files directly.** No converting to TIFF first. Each sub-image inside a LIF becomes its own entry; you can filter by name pattern (e.g. `*Merged*`) to skip raw stacks you don't need.
- **Each channel renders independently.** A 4-channel image (3 fluorophores + brightfield) shows each channel as its own layer with its own LUT, contrast, gamma, opacity, and blend mode — you don't have to squeeze them into R/G/B slots.
- **Big images don't choke the browser.** Past 15 megapixels, Lysa switches to a tiled pyramid renderer (the same approach used by online slide-scanner viewers) and panning a 1.3 GB LIF stays smooth.
- **Measurements feel like measurements, not exports.** Distance, angle, line profile, ROI statistics, multi-point counting — drawn on the image, saved with the session, exported to Excel when you're done.

> **Who this is for.** Biologists working with confocal / wide-field microscopy data who want a fast, repeatable way to view, annotate, and pull numbers out of multi-channel images — without writing scripts and without losing precision.

2. Installation & Running

Requirements

- Python 3.9 or later
- A modern browser with WebGL 2 (Chrome, Firefox, Safari, Edge — all current versions qualify)
- Free disk space roughly equal to your largest LIF file (uploads are cached and content-addressed, so duplicate uploads of the same file don't accumulate)

First-time setup

```
git clone https://github.com/prshanthramachandran/Lysa.git
cd Lysa
pip install -r requirements.txt
```

Running

```
python server.py
```

Then open `http://localhost:8050`. The terminal will show `Application startup complete` when it's ready.

To stop the server: `Ctrl` + `C` in the terminal.

> **Tip.** Lysa uses uvicorn's auto-reload, so editing the source code while the server is running will pick up the changes automatically — handy if you're customizing it.

3. The Interface

The window splits into four regions:

REGION	PURPOSE
Left sidebar	Image library — every image you've loaded, with thumbnails. Drop zones for files and folders. LIF loading options. Session save/load and PDF export.
Top toolbar	Parallel-view count (how many cells in the grid), auto/manual layout, the Tiled toggle, Merge mode, and the Open Images button.
Center viewer	The image grid. One or more cells, each showing one image, sharing pan/zoom when in Auto layout. The currently active cell has a cyan outline.
Right panels	Tools, channel/contrast adjustments, transforms, analysis features, histogram, image info. All collapsible — click a section header to expand.

Below the viewer is a **status bar** showing the cursor's pixel coordinates, the RGB(A) value at that pixel, the image dimensions, dtype, value range, filename, and current zoom level.

The right-hand panels at a glance

PANEL	WHAT IT DOES
Tools	Tool palette — pan, ROIs, measurements, annotations, crop. See the next section.
Channels	The main per-channel control: each source channel becomes a row with its own LUT, contrast sliders, gamma, opacity, visibility, and blend mode.
Contrast Limits	Global window/level (used when there's just one channel, or as a coarse override).
Colormap / LUT	Single-channel colormap (viridis, magma, etc.) plus invert toggle.
Display	Brightness, opacity, attenuation — light cosmetic adjustments that don't affect saved measurements.
Transform	Rotate, flip, crop. Modifies the underlying pixels (the result becomes the new "current" image; reload to undo).
Filters & Preprocessing	Gaussian, median, sharpen, contrast-stretch — applied server-side, replaces the active image.
Thresholding	Otsu (automatic) or manual threshold. Live preview as a colored overlay; "Apply" bakes it into a binary mask.
Segmentation	Watershed-style splitting of thresholded blobs into individual objects, with per-object stats.
Root Growth	Multi-step workflow for plant root growth experiments — define plates, genotype boxes, click root tips across timepoints.
Annotations	Visibility toggle, default color/width/font, save/load JSON.
Batch Analysis	Apply the active image's ROI / adjustments / threshold / crop to every open image at once.
Histogram	Live per-channel intensity histogram.
Image Info	Filename, dimensions, dtype, channels, file size, ingest stats, pixel size if known.
Scale Bar	Pixel-size readout used by the scale bar overlay; auto-detected from TIFF metadata or set via the Calibrate tool.

4. Opening Images

Drag and drop

Drop a single image file or a folder onto the left-sidebar drop zones. Folders load every supported file inside (PNG, JPG, TIFF, BMP, GIF, LIF).

The Open Images button

Top-right of the toolbar — opens a native file picker.

LIF files

Leica LIF containers hold many sub-images per file (different acquisitions, channels, time points). When you drop a LIF, you get three options under **"When dropping a LIF"** in the sidebar:

`Load all`	Every sub-image becomes its own library entry.
`Merged only`	Only sub-images whose name contains `Merged` — usually the channel-merged composites and not the per-channel raws.
`Custom`	A glob pattern you type into the box. `*Col_3*` filters to a single column; `*t6*` filters to a timepoint.

Each loaded sub-image gets a name like `20260506_MfnG_lines_roottip_t6/Col_2_Merged` in the sidebar — the part before the slash is the LIF filename, after it is the sub-image label.

The image library

Each image in the sidebar shows a thumbnail, the filename (truncated), and metadata (dimensions, mode, file size). The **eye icon** toggles whether that image appears in the grid view. The **checkbox** selects images for batch operations.

Click anywhere else on the card to open that image in the active grid cell.

Supported formats

FORMAT	NOTES
`.png` `.jpg` `.bmp` `.gif`	Standard formats. Treated as 8-bit unless otherwise tagged.
`.tif` `.tiff`	Multi-bit-depth. Preserves 16-bit precision. Reads ImageJ pixel-size tags if present.
`.lif`	Leica LAS X. Multi-image, multi-channel, with metadata for channel naming.

5. Drawing & Measurement Tools

Tools live in the **Tools** panel (top of the right sidebar). Click an icon, or use the keyboard shortcut shown in the icon's tooltip. Each tool has a single-letter shortcut.

While a tool is active, the cursor becomes a crosshair over the image. Pan tool is the default; switch to it (`H`) when you just want to move around without drawing.

ROI tools (regions of interest)

TOOL	SHORTCUT	HOW TO USE
Rectangle ROI	`R`	Click and drag to define a rectangle.
Ellipse ROI	`E`	Click and drag to define the bounding box of an ellipse.
Polygon ROI	`G`	Click each vertex; double-click (or click the first vertex) to close.

Saved ROIs appear in the **ROI Measurements** panel below the tool palette: name, area in px and physical units (if calibrated), mean / min / max / std-dev across each channel. The eye icon toggles visibility on the canvas; the trash icon deletes.

Point and line tools

TOOL	SHORTCUT	HOW TO USE
Multi-Point	`P`	Click to add labelled points (cell counting, landmark marking). Double-click to finalize the point set.
Line Profile	`L`	Click waypoints to build a polyline; double-click to finish. Plots intensity along the line.
Freehand	`F`	Press, drag a continuous curve, release to finish.

Measurement tools

TOOL	SHORTCUT	HOW TO USE
Distance	`M`	Click two points. The distance label appears mid-line, in pixels and physical units (if calibrated).
Angle	`A`	Click the vertex first, then the two arm endpoints. Saved as soon as the third click lands.
Calibrate	`C`	Draw a line of known length, then enter the real-world distance. Sets the image's pixel size for all subsequent measurements.

Crop tools

TOOL	HOW TO USE
Crop (drag)	Drag a rectangle. A confirm/cancel popover appears.
Crop (4-corner)	Click 4 corners — useful when the area you want isn't axis-aligned.

Both replace the image with the cropped result. Reload the image (sidebar) to undo.

Annotations

Annotations are decorative — they ride on top of the image but aren't measurements. Use them for figure prep.

TOOL	SHORTCUT	HOW TO USE
Text	`T`	Click to place; type the text in the prompt.
Arrow	`W`	Drag from tail to head.
Scribble	`S`	Press and drag — freehand pen.
Annotation Rectangle	`⇧ R`	Decorative box (no measurement saved).
Annotation Ellipse	`⇧ E`	Decorative ellipse.

The **Annotations** panel sets the default color, line width, and font. Existing annotations stay at the color they were drawn with.

6. Channels, Contrast & LUTs

For multi-channel images, the **Channels** panel is where you'll spend most of your display-tweaking time.

One row per source channel

Each row controls one channel of the source image, independently of the others. You can:

- **Toggle visibility** with the eye icon
- **Pick a LUT** from the dropdown — Gray, Red, Green, Blue, Cyan, Magenta, Yellow, plus colormaps (Viridis, Magma, Inferno, Hot, Cool, Turbo)
- **Drag the contrast Min / Max sliders** in the channel's native units (e.g. 0 to 65535 for 16-bit)
- **Set Gamma** from 0.1 to 5.0
- **Set Opacity** from 0 to 1
- **Pick a Blend mode**: Additive (sum), Max (brightest channel wins), or Over (alpha blend on top)

Auto-contrast on load

When you open an image, each channel's Min/Max defaults to its own 1st–99th percentile — the same stretch the thumbnail uses. Bright channels won't look saturated and dim channels won't look invisible. You can re-apply this any time with the **Auto** button in each channel row.

BUTTON	ACTION
Auto	Set Min/Max to this channel's 1/99 percentile (best general-purpose default).
Min/Max	Stretch to the channel's actual min and max. Sensitive to outlier pixels.
Full	Use the full dtype range (0..255 for 8-bit, 0..65535 for 16-bit). Often very dim — useful as a reset.

Default channel colors

Lysa picks defaults based on what kind of image it is:

IMAGE TYPE	DEFAULT CHANNEL COLORS
Stored RGB / RGBA	Channel 0 → Red, channel 1 → Green, channel 2 → Blue. Matches the file's natural meaning.
Fluorescence (LIF, multi-channel TIFF)	Channel 0 → Green (typical 1st fluorophore), 1 → Red, 2 → Blue, 3 → Gray (brightfield), then Magenta, Yellow, Cyan rotated.

You can change any channel's color at any time from its row's dropdown.

> **Why this matters.** A GFP image saved as a 2-channel "RGB" TIFF actually has the green signal in channel 1 and channel 0 empty. Before Lysa got the RGB-mode detection right, the default was the fluorescence convention, so channel 1 (the green signal) ended up rendered with the red LUT — blown-out red specimen on a black field. Now stored-RGB images render the way their thumbnail looks.

Reset

The **Reset All** link in the Channels panel header clears all customization for the active image and re-applies the defaults.

7. Tiled Mode (Large Images)

Loading a 1.3 GB LIF as a single canvas would chew through browser memory and freeze the tab. Lysa avoids this by switching to a **tiled pyramid renderer** — the image is broken into 512×512-pixel tiles at multiple resolutions, and only the tiles needed for the current view are decoded.

When does it kick in?

- **Automatically** for images at or above the megapixel threshold set in the sidebar (default: 15 MP). Adjust with the input under **"Auto-enable Tiled mode for images ≥"**.
- **Manually** via the **Tiled** toggle button in the top toolbar — forces every open image into tiled mode.

What's different in tiled mode?

- Pan and zoom feel like Google Maps — smooth, asynchronous tile loading.
- Each channel renders into its own background OpenSeadragon viewer, then a WebGL2 shader composites them according to your LUT/contrast/blend settings. You don't see the per-channel viewers; what you see is the composite.
- The full bit-depth raw data isn't fetched (it would exhaust browser memory for a 1.3 GB file). The "Auto" button reads its 1/99 percentiles from the server-side pyramid instead — same numbers, same outcome.
- The **Filters & Preprocessing** and **Transform** panels still work — the server applies the operation, then the tile pyramid is rebuilt.

Threshold for switching

The default 15 MP threshold is conservative — even modern laptops handle 15 MP images fine in legacy mode. Raise it (e.g. 30 MP) if you have lots of RAM and want full 16-bit-precision raw data for measurements; lower it if your machine struggles with smaller images.

8. Analysis Features

Thresholding

Open the **Thresholding** panel. Pick a method (**Otsu** for automatic, **Manual** for a slider) and a value — the threshold preview overlays on the image as a colored mask. The histogram below shows the threshold position relative to the intensity distribution.

Apply on all runs the same threshold method against every open image — useful for batch quantification.

Segmentation

Once a threshold is set, Segmentation can split the binary mask into individual objects (watershed-style) and report per-object statistics — area, centroid, intensity in each channel. The per-object table appears below the panel; click an object's row to highlight it on the image.

Root Growth

A specialized multi-step workflow for plant root experiments. The panel walks you through:

1. **Define plates** — drag rectangles around each agar plate
2. **Define genotype boxes** — drag rectangles around each genotype's seedlings within the plates
3. **Click root tips** at each timepoint — Lysa records labelled coordinates and computes growth rates

Filters & Preprocessing

Server-side image filters: Gaussian blur, median, unsharp mask, contrast stretch, and a few others. Each replaces the active image with the filtered result. The **Reset** link in the panel header restores the original.

Histogram

Live per-channel intensity histogram of the active image. Updates as you adjust contrast.

9. Sessions, Batch & Export

Save / Load Session

The **Save Session** button (left sidebar) writes a JSON file containing:

- The list of currently open images (paths, not pixel data)
- All saved ROIs, polylines, points, measurements, and annotations per image
- Per-image channel settings (LUTs, contrast windows, gamma, opacity, blend modes)
- UI state: parallel-view count, tiled mode, active tab

Load Session restores everything. The image files themselves need to still exist at their original paths — sessions reference, they don't archive.

Batch Analysis

The **Batch Analysis** panel takes the operation you set up on the active image and applies it to every other open image. Buttons:

Apply last ROI to all	Copies the most recently drawn ROI onto every open image at the same coordinates. Useful when all images are the same field of view (e.g., timepoints).
Apply current adjustments	Copies the active image's contrast / LUT / channel settings to every open image.
Run threshold on all	Applies the active threshold method/value to every open image.
Apply last crop to all	Copies the same crop rectangle to every open image.
Export Everything	Writes the per-image stats plus histograms and charts to one Excel workbook.

Export PDF

Bottom-left button. Renders all open images plus their ROI tables, measurements, and histograms into a single PDF — good for figure prep or supplementary material.

10. Cheatsheet

Keyboard shortcuts

KEY	ACTION
H	Pan tool
R	Rectangle ROI
E	Ellipse ROI
G	Polygon ROI (double-click to close)
P	Multi-Point
L	Line Profile (double-click to finish)
M	Distance Measure
A	Angle (vertex first, then arm endpoints)
F	Freehand
C	Calibrate (set pixel size)
T	Text annotation
W	Arrow annotation
S	Scribble annotation
⇧ R	Annotation Rectangle
⇧ E	Annotation Ellipse
+ / =	Zoom in
−	Zoom out
0	Fit to view

Common workflows

QUICK-LOOK AT A LIF

1. Drop the LIF into the sidebar; choose **Merged only** if you don't need the per-channel raws.
2. Click any sub-image card to open it. Tiled mode auto-engages above 15 MP.
3. Use the **Channels** panel to fine-tune contrast per channel. Auto button if defaults look off.

MEASURE 20 ROOT TIPS ACROSS TIMEPOINTS

1. Open all timepoint images (drop the folder).
2. Set parallel views to a number that fits — 4 or 6 works well — and pick **Multi-Point** (`P`).
3. Click each root tip per image. Lysa records labelled coordinates per image.
4. **Export Everything** for an Excel summary.

CALIBRATE THE SCALE BAR

1. Select **Calibrate** (`C`).
2. Draw a line on something of known length (e.g. a stage micrometer line, or a feature you've measured before).
3. Enter the real-world length when prompted. Pixel size now propagates to all measurements and the scale bar overlay.

APPLY THE SAME ROI TO A SERIES OF IMAGES

1. Open all images in the series.
2. Draw the ROI on the first one.
3. In **Batch Analysis**, click **Apply last ROI to all**. Stats are computed for each image at the same coordinates.

Troubleshooting

SYMPTOM	LIKELY CAUSE / FIX
Image renders blown-out on first load	Click **Auto** on the affected channel(s). If this keeps happening, the image's metadata may be missing percentile stats — re-upload to recompute.
Tools don't respond to clicks	Make sure a tool is selected (not pan). The cell border should be cyan with a crosshair cursor over the image. Refresh the browser if it persists.
500 errors when panning a tiled image	Backend hasn't finished building the pyramid for that channel. Wait a few seconds and pan again.
`uploads/` folder is huge	Lysa caches uploaded files and dedupes by content hash, but each unique file is kept. Manually delete files from `uploads/` when you're done with a session — Lysa will re-cache on next upload.
LIF sub-images load slowly	Use the **Merged only** or **Custom *pattern*** option to skip sub-images you don't need. Each sub-image still costs disk and RAM if loaded.

Tools at-a-glance

H Pan

P Multi-Point

R Rectangle ROI

L Line Profile

E Ellipse ROI

M Distance

G Polygon ROI

A Angle

F Freehand	**C** Calibrate
T Text	**W** Arrow
S Scribble	⇧ **R** Ann. Rect
⇧ **E** Ann. Ellipse	